

Name:		 MANIPAL UNIVERSITY JAIPUR		
Enrolment No:				
Odd Semester End Term Examination, November 2025 Faculty of Science, Technology and Architecture School of Physical and Biological Sciences Department of Chemistry BTech 1 st Year				
Course Code: CHY1001/CY1001		Course: Engineering Chemistry	Semester: I	
Time: 3 hrs.			Max. Marks: 80	
Instructions: All questions are compulsory. Missing data, if any, may be assumed suitably. A non-programmable calculator is allowed.				
S.N.	SECTION-A		Marks	CO
Q1	Which method is used to determine the hardness of water using EDTA? (1) Complexometric titration (2) Gravimetric analysis (3) pH metric titration (4) Colorimetric analysis		[2]	1
Q2	A polymer has a number-average molecular weight (M _n) of 30,000 g/mol. If the molecular weight of the repeating unit is 120 g/mol, the degree of polymerization (DP). (1) 250 (2) 2500 (3) 360 (4) 3600		[2]	2
Q3	Which of the following methods is commonly used to determine the nitrogen content in coal during ultimate analysis? (1) Boy's calorimetry (2) Bomb calorimetry (3) Kjeldahl's method (4) Proximate analysis		[2]	3
Q4	Which of the following factors related to the nature of the metal can significantly affect its corrosion rate? (1) Density (2) Melting point (3) Position in the electrochemical series (4) Thermal conductivity		[2]	4
Q5	Which of the following monomers is least likely to undergo anionic polymerization? (1) Acrylonitrile (2) Methyl methacrylate (3) Styrene (4) Isobutylene		[2]	2
S.N.	SECTION-B		Marks	CO
Q6	Calculate the temporary hardness and permanent hardness of a sample of water containing: Mg(HCO ₃) ₂ = 7.3 mg/L; Ca(HCO ₃) ₂ = 16.2 mg/L; MgCl ₂ = 9.5 mg/L; CaSO ₄ = 13.6 mg/L. [Atomic Mass: Ca = 40, Mg = 24, Na = 23, C = 12, O = 16, H = 1, Cl = 35.5, S = 32]		[6]	1
Q7	Apply the fixed bed catalytic method for the cracking of heavy oil and describe the process with a diagram.		[6]	3
Q8	Explain the processes of internal treatment methods of hard water with chemical reactions.		[6]	5
Q9	Describe the polymerization of vinyl chloride (CH ₂ =CH-Cl) in presence of potassium amide.		[6]	2
Q10	Illustrate the electrochemical corrosion mechanism of iron with cathodic and anodic reactions in water and acidic medium.		[6]	4

S.N.	SECTION-C	Marks	CO																									
Q11	(i) A 2 g air-dried coal sample was oven dried at 110 °C for an hour and the loss in weight was 0.15 g. Then the sample was heated in an oxygen-less environment at 950 °C for 7 minutes and the sample weighed 1.02 g. Then the remaining sample was ignited at 700 °C till a constant weight of 0.30 g was obtained. Based on this information, identify what kind of analysis this is and what conclusion you can draw from it. (ii) Synthesize the Nylon 6,10 using proper reagents and predict the final polymeric structure.	[7+3]	3, 2																									
Q12	(i) Justify the following cases: (a) A screw and nut made of the same metal are preferred in practice to avoid corrosion. (b)The part of a metal sheet covered with a piece of wood undergoes more corrosion compared to the part freely exposed to air. (c) Iron does not corrode when the coating is broken in a galvanized iron pipe. (d) Zinc corrodes under drops of salt solution. (ii) 3 kg of a fuel sample contains 10 % of hydrogen upon burning in a calorimeter, liberating 15000 kcal of heat. Calculate the gross and net calorific value of the fuel.	[8+2]	3,4																									
S.N.	SECTION-D	Marks	CO																									
Q13	(i) Calculate the Pilling Bedworth (P-B) ratio of the following metal oxides and find out which of the metal oxide coatings will be protective/nonprotective in nature? <table><thead><tr><th>Metal Oxide</th><th>Atomic weight of metal</th><th>Molecular weight of Metal oxide</th><th>Metal oxide density (g/ cm³)</th><th>Metal density (g/ cm³)</th></tr></thead><tbody><tr><td>NiO</td><td>59</td><td>75</td><td>6.67</td><td>8.90</td></tr><tr><td>MgO</td><td>24</td><td>40</td><td>3.58</td><td>1.74</td></tr><tr><td>CaO</td><td>40</td><td>56</td><td>3.34</td><td>1.55</td></tr><tr><td>CuO</td><td>63.5</td><td>79.5</td><td>6.31</td><td>8.96</td></tr></tbody></table> (ii) Calculate how many moles of soda are required for the following salts: (a) MgSO₄ (b) Mg(HCO₃)₂ (c) CaCl₂ (d) CaSO₄	Metal Oxide	Atomic weight of metal	Molecular weight of Metal oxide	Metal oxide density (g/ cm ³)	Metal density (g/ cm ³)	NiO	59	75	6.67	8.90	MgO	24	40	3.58	1.74	CaO	40	56	3.34	1.55	CuO	63.5	79.5	6.31	8.96	[8+2]	1, 4
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Q14	(i) Consider a polymer sample containing 50 molecules of molecular weight 10000, 35 molecules of molecular weight 12000, and 15 molecules of molecular weight 14000. Determine number average molecular weight, weight average molecular weight. If the polymer is polypropylene, what will be the degree of polymerization? (ii) Choose a suitable calorimeter to calculate calorific value of non-volatile fuel and describe its working principle with diagram.	[5+5]	2,3																									

QUESTION PAPER MODERATOR'S CHECKLIST

Programme: B-Tech 1 ST Year		Semester: II	Session: 25-26
Course Code: CHY1001/CY1001	Course Title: Engineering Chemistry		
Exam Type (MTE / ETE / Make-Up): ETE			
Approx. No. of students appearing for the exam: 1800+			

Suggested weightage of Section wise Marks:

Section - A	Section - B	Section - C	Section - D
BL1-Remember	BL2-Understand & BL3-Apply	BL4-Analyse & BL5-Evaluate	BL6-Create
MTE Marks - 6	MTE Marks - 16	MTE Marks - 8	NA
ETE Marks - 10	ETE Marks - 30	ETE Marks - 20	ETE Marks - 20

BL: Bloom's Level

Unit/Topic wise Distribution and weightage of marks

Unit	Actual marks in Question paper (marks to be filled below)								Total Marks	Total Question Count
	Section - A		Section - B		Section - C		Section - D			
	Marks	Question Count	Marks	Question Count	Marks	Question Count	Marks	Question Count		
Unit 1	2	1	12	2	0	0	5	1	19	4
Unit 2	4	2	6	1	6	2	2	1	18	6
Unit 3	2	1	6	1	8	1	5	1	21	4
Unit 4	2	1	6	1	6	1	8	1	22	4
Total Marks (MTE-30/ ETE-80)	10	5	30	5	20	4	20	4	80	18

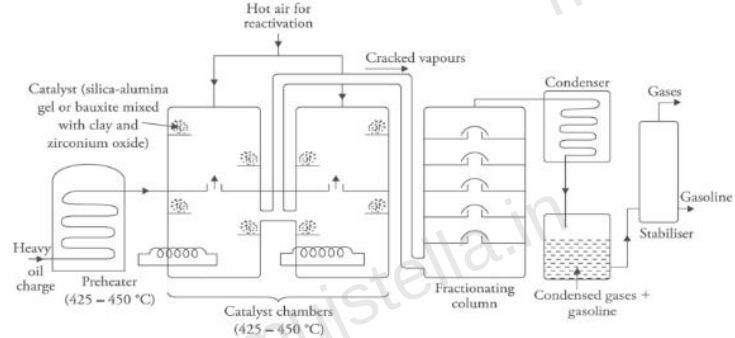
Note: 60% of the course content to be covered in MTE and 100% in ETE

Summary Checklist:

Sl. No	Checklist	Yes/No	Remarks
1	Do the questions address the objectives/topics within the recommended syllabus?	yes	-----
2	Is the distribution of marks close to the recommended marks? Ensure that the actual marks in question paper are in the range of $\pm 20\%$ marks assigned to the topic.	yes	-----
3	Are the questions framed as a sentence and structured with appropriate action verbs of Blooms Taxonomy and clearly delimited?	yes	-----
4	Are the questions answerable within the allotted time limit?	yes	-----

(1) Name and Signature of the Paper Setter **Dr. Suranjan De**

(2) Name and Signature of the Paper Moderator (HoD/HoD nominee) **Dr. Praveen Kumar Surolia**

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Course Code: CHY1001/CY1001		Course: Engineering Chemistry	Semester: I
Time: 3 hrs.		Max. Marks: 80	
Answer Scheme			
S.N.	SECTION-A	Marks	CO
Q1	(1) Complexometric titration	[2]	1
Q2	(1) 250	[2]	2
Q3	(3) Kjeldahl's method	[2]	3
Q4	(3) Position in the electrochemical series	[2]	4
Q5	(4) Isobutylene	[2]	2
S.N.	SECTION-B	Marks	CO
Q6	CaCO ₃ Equivalent: Mg (HCO ₃) ₂ : = 5.0 mg/L ; Ca (HCO ₃) ₂ : = 10.0 mg/L , MgCl ₂ : mmol = 10.0 mg/L ; CaSO ₄ : mmol = 10.0 mg/L [4 X 1 = 4 Marks] Temporary hardness = 5.0 + 10.0 = 15.0 mg/L as CaCO₃ [1 Mark] Permanent hardness = 10.0 + 10.0 = 20.0 mg/L as CaCO₃ [1 Mark]	[6]	1
Q7	Fixed bed catalytic cracking – Heavy oil is pre-heated to 425–450 °C and passed through a catalytic chamber with artificial clay and zirconium oxide at 1.5 kg/cm ² . About 40% converts to gasoline and 2–4% carbon forms. Vapors are fractionated, cooled, condensed, and finally sent to a stabilizer to remove gases and obtain pure gasoline. [4 Marks] ; Proper Diagram [2 Marks]  Figure 1.11 Fixed bed catalytic cracking	[6]	3
Q8	Phosphate conditioning [1 Mark] + Reaction with salt [1 Mark] $3\text{CaCl}_2 + 2\text{Na}_3\text{PO}_4 \rightarrow \text{Ca}_3(\text{PO}_4)_2 + 6\text{NaCl}$ Calgon conditioning [1 Mark] + Reaction with salt [1 Mark] $2\text{CaSO}_4 + [\text{Na}_4\text{P}_6\text{O}_{18}]^{2-} \rightarrow [\text{Ca}_2\text{P}_6\text{O}_{18}]^{2+} + 2\text{Na}_2\text{SO}_4$ Colloidal treatment [1 Mark], + reagents (glue, seaweed extract, agar-agar, tannin) [1 Mark]	[6]	5

Q9	① <u>Initiation</u>: $K^+ \text{NH}_2^- + \text{CH}_2=\text{CH}-\text{Cl} \longrightarrow \text{H}_2\text{N}-\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}\text{K}^+ \quad [2 \text{ Marks}]$ ② <u>propagation</u>: $\text{NH}_2-\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}\text{K}^+ + n \text{CH}_2=\text{CH}-\text{Cl} \longrightarrow \text{NH}_2-\left[\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}\right]_n-\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}\text{K}^+ \quad [2 \text{ Marks}]$ ③ <u>Termination</u>: $\text{NH}_2-\left[\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}\right]_n-\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}\text{K}^+ \xrightarrow{\text{H}^+} \text{NH}_2-\left[\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}\right]_n-\text{CH}_2-\overset{(-)}{\underset{\text{Cl}}{\text{CH}}}-\text{H} \quad [2 \text{ Marks}]$ [No Marks for wrong mechanism or wrong steps or wrong structures]	[6]	2
Q10	Illustrate the electrochemical corrosion mechanism of iron with cathodic and anodic reactions in water and acidic medium. Anodic reaction: $\text{Fe (s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$ [1 Mark] Cathodic reaction: a) Evolution of H₂ (acidic medium) in absence of oxygen [1 Mark] Cathode: $2 \text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2$ b) Absorption of O₂ : Neutral medium [1 Mark] Cathode: $\frac{1}{2} \text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{aq}) + 2\text{e}^- \rightarrow 2\text{OH}^-$ (c) Deaerated and neutral medium [1 Mark] Cathode: $\text{H}_2\text{O}(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ Final reaction: [1+1=2 Marks] $\text{Fe}^{2+} + 4 \text{OH}^- \longrightarrow 2\text{Fe}(\text{OH})_2$ $2\text{Fe}(\text{OH})_2 + \frac{1}{2} \text{O}_2 + (\text{x}-2) \text{H}_2\text{O} \longrightarrow \text{Fe}_2\text{O}_3 \cdot \text{xH}_2\text{O}$	[6]	4
S.N.	SECTION-C	Marks	CO
Q11	(i) Proximate analysis: [2 Marks] Conclusions: Moisture= 7.5 % [1 Mark]; Volatile material= 41.5 % [1 Mark], Ash = 15 % [1 Mark], Fixed carbon= 36 % [1 Mark] (ii) Proper reagent name/Structure: $\left[\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \quad \text{O} \\ \quad \quad \quad \\ \text{N}-(\text{CH}_2)_6-\text{N}-\text{C}-(\text{CH}_2)_8-\text{C} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{Cl} \quad \text{Cl} \end{array} \right]_n$ Hexamethylene diamine Decanedioyl dichloride Polymer Final Structure: [1 Mark] $\left[\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \quad \text{O} \\ \quad \quad \quad \\ \text{N}-(\text{CH}_2)_6-\text{N}-\text{C}-(\text{CH}_2)_8-\text{C} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{Cl} \quad \text{Cl} \end{array} \right]_n$ Nylon 6:10	[7+3]	3, 2
Q12	(i) (a) To prevent galvanic (bi-metallic) corrosion [2 marks] (b) Differential aeration corrosion, less oxygenated part (inside wood) acts as anode. Corrosion occurs at anodic region. [2 marks] (c) In galvanized iron pipe, Zinc is coated over Fe and since Zn is more active (anodic) as per the electrochemical series, it acts as anode and Fe act as cathode and corrosion happens on anode only. [2 marks] (d) Differential aeration corrosion, less oxygenated part of Zn (under the water droplet) acts as anode. Corrosion occurs at anodic region. [2 marks]	[8+2]	3,4

	(ii) Gross Calorific value = $\frac{15000}{3} \text{ kcal/kg} = 5000 \text{ kcal/kg}$ [1 Mark] Net Calorific value = $5000 - (0.09 \times 10 \times 587) = 4471.7 \text{ kcal/kg}$ [1 Mark]																						
S.N.	SECTION-D	Marks	CO																				
Q13	(i) $R_{PB} = \frac{V_{\text{Oxide}}}{V_{\text{Metal}}} = \frac{M_{\text{Oxide}} \times \rho_{\text{Metal}}}{n \times M_{\text{Metal}} \times \rho_{\text{Oxide}}}$ (only formula 1 Mark if math is wrong) <table border="1"> <thead> <tr> <th>Metal Oxide</th><th>PB Ratio (1 Mark)</th><th>Nature (1 Mark)</th><th>Total Marks</th></tr> </thead> <tbody> <tr> <td>NiO</td><td>1.696</td><td>protective</td><td>2</td></tr> <tr> <td>MgO</td><td>0.81</td><td>nonprotective</td><td>2</td></tr> <tr> <td>CaO</td><td>0.649</td><td>nonprotective</td><td>2</td></tr> <tr> <td>CuO</td><td>1.77</td><td>protective</td><td>2</td></tr> </tbody> </table> (ii) (a) 1 mole (b) Not Required/NA (c) 1 mole (d) 1 mole [0.5*4 =2 Marks] $\text{MgSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{MgCO}_3 \downarrow + \text{Na}_2\text{SO}_4$ $\text{Mg}(\text{HCO}_3)_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{No Reaction/ NA}$ $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 \downarrow + 2\text{NaCl}$ $\text{CaSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 \downarrow + \text{Na}_2\text{SO}_4$ If proper balanced reactions written considered marks	Metal Oxide	PB Ratio (1 Mark)	Nature (1 Mark)	Total Marks	NiO	1.696	protective	2	MgO	0.81	nonprotective	2	CaO	0.649	nonprotective	2	CuO	1.77	protective	2	[8+2]	1, 4
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CuO	1.77	protective	2																				
Q14	(i) Number average MW (M_n) = 11300 [2 Marks] $\overline{M}_n = \frac{1,130,000}{100} = 11300$ Weight average MW (M_w) = 11486.725 [2 Marks] $\overline{M}_w = \frac{12.98 \times 10^9}{1.13 \times 10^6} = 11486.725$ Degree of polymerization (PP) ≈ 269 [1 Mark] $\text{DP} = \frac{M_n}{42} = \frac{11300}{42} = 268.9 \approx 269$ (ii) Choose a suitable calorimeter: Bomb Calorimeter [1 Mark]. Principle: Heat liberated by the fuel = Heat gained by water and calorimeter [1 Mark] Description of calorimeter diagram [2 Marks]; $m\theta = (W + w)(t_2 - t_1) S \quad (\text{cal})$ GCV/HCV [0.5 Mark]: $\theta = \frac{(W+w)(t_2-t_1)S}{m} \quad (\text{cal/g})$ LCV/NCV formula [0.5 Mark]: $= \text{HCV} - 0.09H \times 587 \text{ cal (in percentage)}$	[5+5]	2,3																				